

Bureau of Indian Standards

Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 2185:2005

2005

CONCRETE MASONRY UNITS - SPECIFICATION

SPECIFICATION

ICS 91.100.30

BIS 2005

Indian Standard

1 SCOPE

This standard covers the requirements for concrete masonry units (blocks) made from Portland cement, water, and aggregates with or without additions. It specifies requirements for solid blocks, hollow blocks, and light-weight blocks.

2 CLASSIFICATION

Concrete blocks are classified as follows:

- Type A: Load-bearing blocks (for structural walls)
- Type B: Non-load-bearing blocks (for partition walls)
- Type C: Insulating blocks (thermal insulation)

3 DIMENSIONS AND TOLERANCES

Nominal Size (LWH)	Tolerance (L)	Tolerance (W)	Tolerance (H)
390190190 mm	3 mm	3 mm	3 mm
39019090 mm	3 mm	3 mm	3 mm
390290190 mm	3 mm	3 mm	3 mm

4 COMPRESSIVE STRENGTH

Block Type	Average Strength MPa, min	Individual Strength MPa, min
Type A (Load-bearing)	3.5	2.8
Type B (Non-load-bearing)	2.5	2.0
Type C (Insulating)	1.5	1.2

5 WATER ABSORPTION

The water absorption shall not exceed 15% by mass for normal-weight blocks and 20% by mass for light-weight blocks, when tested as per IS 2185.

6 TEST METHODS

- Compressive Strength: IS 2185 Part 1 - Cube Test (150mm)
- Water Absorption: IS 2185 Part 1 - Soaking Method
- Dimensional Tolerance: IS 2185 Part 1 - Measurement
- Drying Shrinkage: IS 2185 Part 1 - 28-day drying test